IEEE WIE ILS 2026 · HackX Grand Finale · SVNIT Surat · 25 October 2026

Track 3: SheLeads · P6: Women's Safety Wearable System

EDGE SAFETY · WEARABLE COMPUTING · HUMAN-CENTRED DESIGN

RESEARCH PROTOTYPE • 2026

NAARI OS

A wrist-first safety system for faster, deliberate help requests

TEAM PURAANMAYA

COLLEGE: _____

1 THE PROBLEM

In a fall or unsafe journey, unlocking and operating a phone can delay a request for help. A wearable must make the action immediate while limiting accidental escalation.

1 action
from the wrist to start an SOS workflow

2 OUR SOLUTION

Naari OS combines a touch-screen ESP32-S3 watch and an Android companion. The watch keeps critical sensing and interaction local; the phone provides guardian communication and location services over authenticated BLE.

DESIGN HYPOTHESIS

- Wrist-first action reduces interaction effort under stress.
- A 30 s review window and 3 s hold reduce accidental cancellation.
- Local sensing continues even when the phone link is unavailable.

3 HOW IT WORKS

1 SIGNALS + USER INPUT
Touch SOS · QMI8658 motion · microphone

↓

2 EDGE DECISION ON WATCH
Fall candidate · manual SOS · 30 s cancel

↓

3 AUTHENTICATED BLE BRIDGE
Bonded Android link · request identity · status

↓

4 PHONE ACTION EXECUTOR
Guardian SMS · call request · phone location

↓

5 GUARDIAN RESPONSE
Action result returns to watch and incident log

PARALLEL EVIDENCE PATH
Voice memo → local storage → verified BLE transfer to

4 PROTOTYPE



Motion monitoring · home + · Saved voice memos

Screens rendered by the production C++ watch UI

5 TECH STACK

WATCH	ESP32-S3 · 410×502 AMOLED
SENSING	QMI8658 IMU · touch input
AUDIO	ES7210 ADC · ES8311 codec
FIRMWARE	C/C++ · ESP-IDF · FreeRTOS
CONNECT	NimBLE · authenticated GATT
ANDROID	Kotlin · Jetpack Compose · Room

6 IMPACT & NEXT

Expected impact

- Quicker access to a deliberate safety workflow from the wrist.

Research validation

DEMONSTRATED	Watch UI, IMU, audio + persistent memos
BUILT	Android APK + BLE sync protocol
ONGOING	Fall accuracy + LTE field tests

NEXT: collect wrist-motion datasets, measure false alarms/hour, and complete supervised field trials.

7 TEAM

Team Puraanmaya

1. _____
2. _____
3. _____
4. _____

SCAN THE LIVE DEMO